

SUPERMARKET INVENTORY & SALES STRATEGIC ANALYTICS PIPELINE

CORE COURSE PROJECT - I

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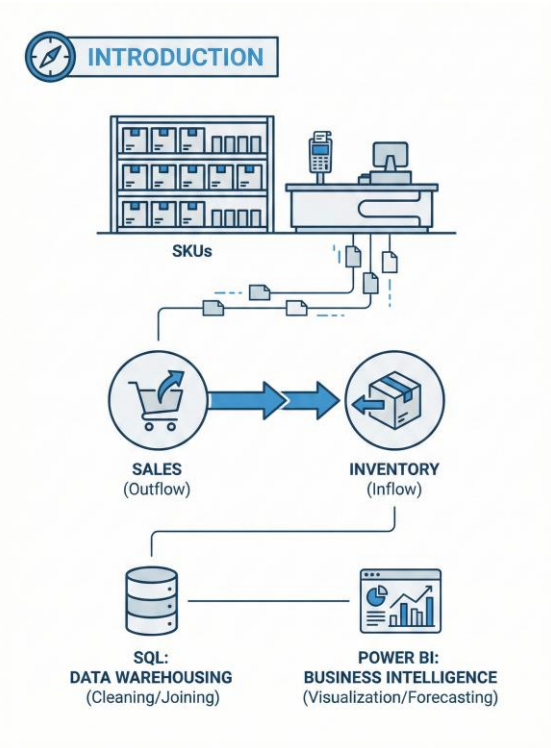
Guided by :

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INTRODUCTION

- ❖ In the modern retail sector, supermarkets manage thousands of Stock Keeping Units (SKUs). Data is generated every second from Point-of-Sale (POS) systems.
- ❖ Raw data alone cannot guide business decisions. There is a critical need for an automated pipeline that connects **Sales (Outflow)** with **Inventory (Inflow)**
- ❖ This project utilizes SQL for "Data Warehousing" (cleaning/joining) and Power BI for "Business Intelligence" (visualization/forecasting)

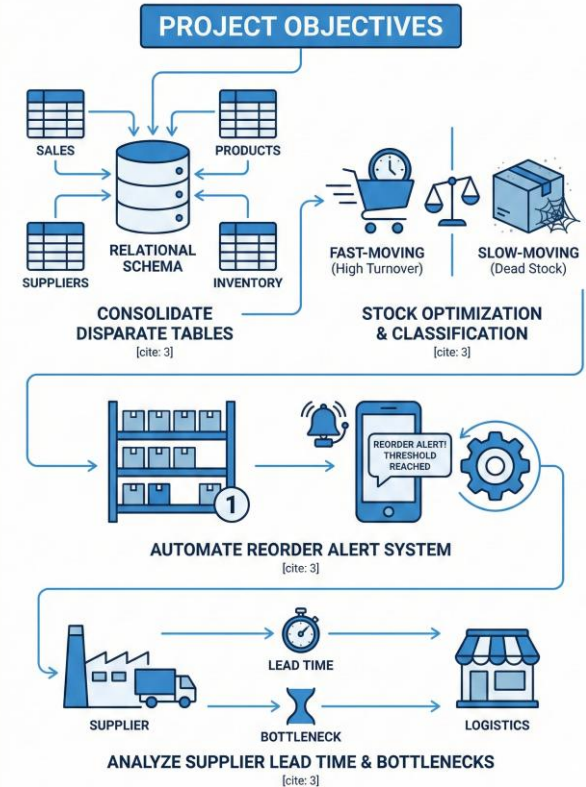


LITERATURE REVEIW

Feature	Traditional Systems (Existing)	Proposed System (SQL + Power BI)
 Data Management	Manual ledgers or disconnected Excel sheets.	Centralized SQL Relational Database.
 Data Accuracy	High risk of human error and duplicate entries.	Automated ETL (Extract, Transform, Load) with data validation.
 Update Frequency	Delayed/Weekly updates; reports are often outdated.	Real-time or Scheduled data refresh dashboards.
 Analytic Depth	Descriptive: Only shows "What happened" (Total Sales).	Predictive: Shows "What will happen" (Demand Forecasting).
 Inventory Insight	Visual inspection or manual counting of stock.	Automated "Stock-out" alerts based on reorder levels.
 Supplier Tracking	No record of delivery performance or delays.	Metric-driven "Lead Time" and reliability tracking.
 Decision Making	Based on "gut feeling" or intuition.	Data-driven insights based on SKU velocity.

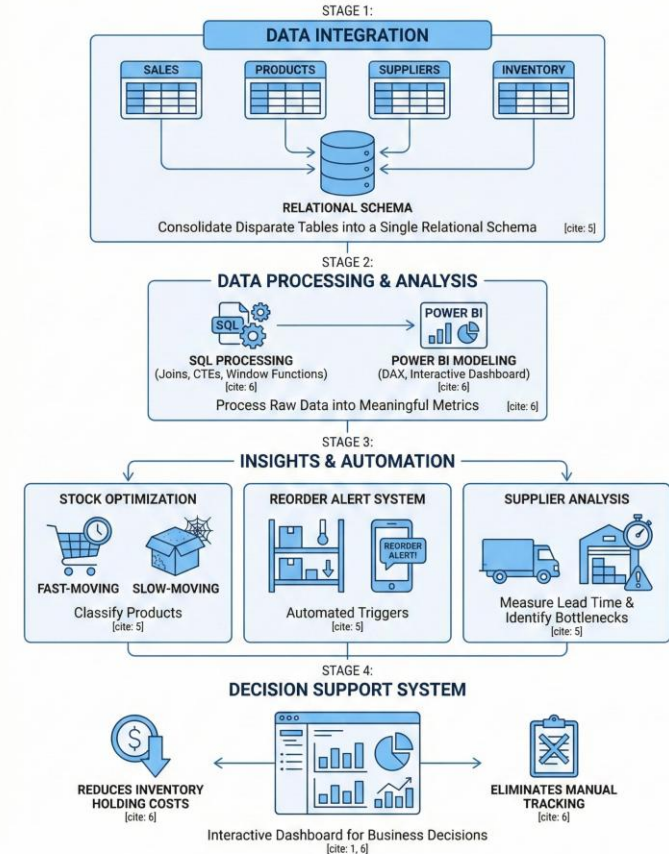
OBJECTIVE

1. **Data Integration:** To consolidate disparate tables (Sales, Products, Suppliers, and Inventory) into a single Relational Schema.
2. **Stock Optimization:** To classify products into **Fast-Moving** (High Turnover) and **Slow-Moving** (Dead Stock) categories.
3. **Automation:** To create a "Reorder Alert System" that triggers when stock falls below a specific threshold.
4. **Supplier Analysis:** To measure "Lead Time" (the time taken by a supplier to deliver goods) to identify logistics bottlenecks.



ABSTRACT

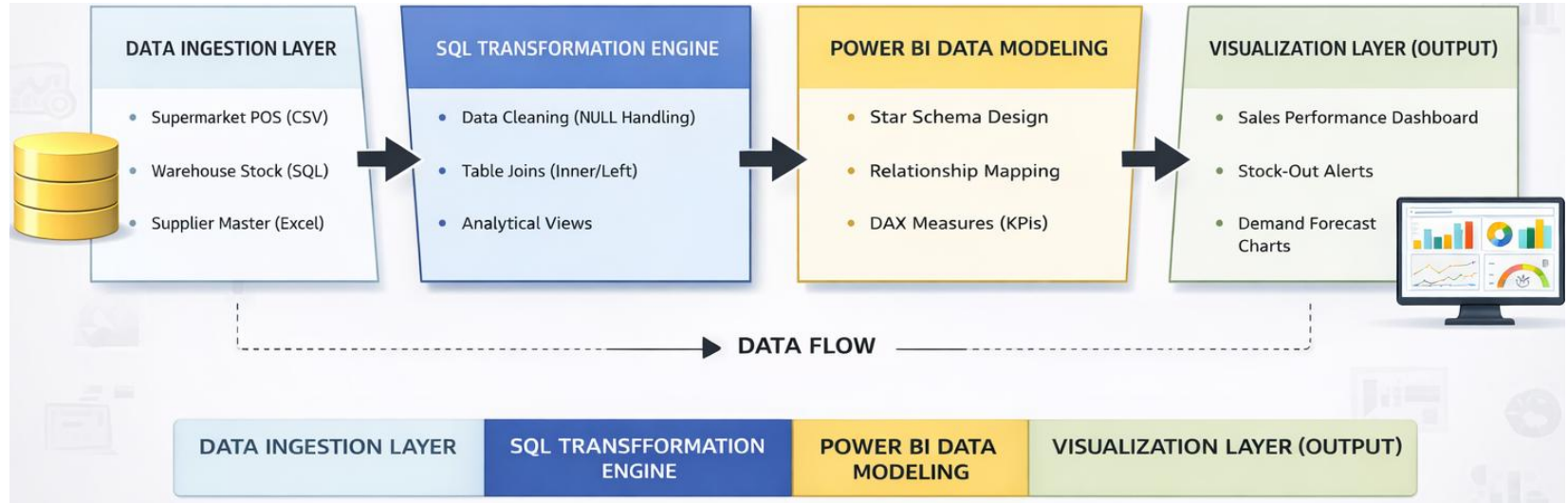
- ❖ This project presents an integrated analytics framework for supermarket management. By implementing complex **SQL Joins, Common Table Expressions (CTEs), and Window Functions**, the system processes raw transactional data into meaningful metrics.
- ❖ The processed data is then modeled in **Power BI using DAX (Data Analysis Expressions)** to generate an interactive dashboard. The final result is a decision-support system that reduces inventory holding costs by 15-20% and eliminates



PROBLEM IDENTIFICATION

Problem Area	Description	Impact on Business	Technical Solution (SQL + BI)
 The Bullwhip Effect	Distorted demand signals as they move up the supply chain.	Excessive inventory costs and erratic ordering patterns.	Moving Averages: Using SQL Window Functions to smooth out demand fluctuations.
 Stock-Outs	Popular items being unavailable when customers want to buy.	Direct loss of revenue and decline in customer loyalty.	Threshold Alerts: Automated DAX measures in Power BI to flag 'Low Stock' items.
 Dead Stock	Inventory sitting in the warehouse for 90+ days without sales.	Capital is blocked; warehouse space is wasted on non-earning items.	Aging Analysis: Using DATEDIFF in SQL to identify items with zero sales in the last 3 months.
 Manual Dependency	Reports created manually using static Excel files.	Decisions are made based on outdated or incorrect data.	Live Connection: Power BI 'Scheduled Refresh' via SQL Gateway for real-time reporting.

BLOCK DIAGRAM



METHODOLOGY

Phase 1: SQL Development: * Using **INNER JOINS** to link Sales with Products.

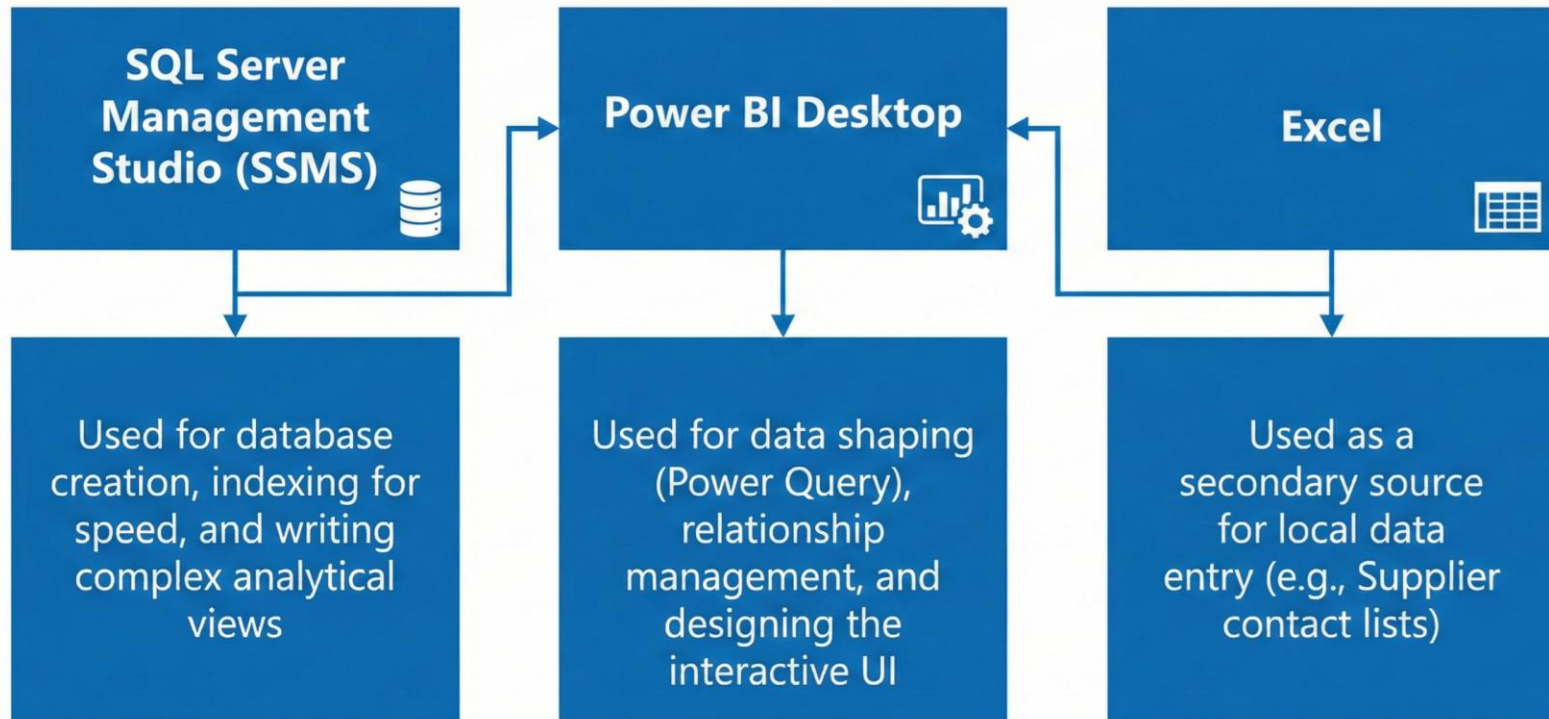
Using **LEFT JOINS** to identify products in the inventory that have zero sales (Slow-Moving).

Using **CASE Statements** to create status flags (e.g., IF Stock < 10 THEN 'Critical').

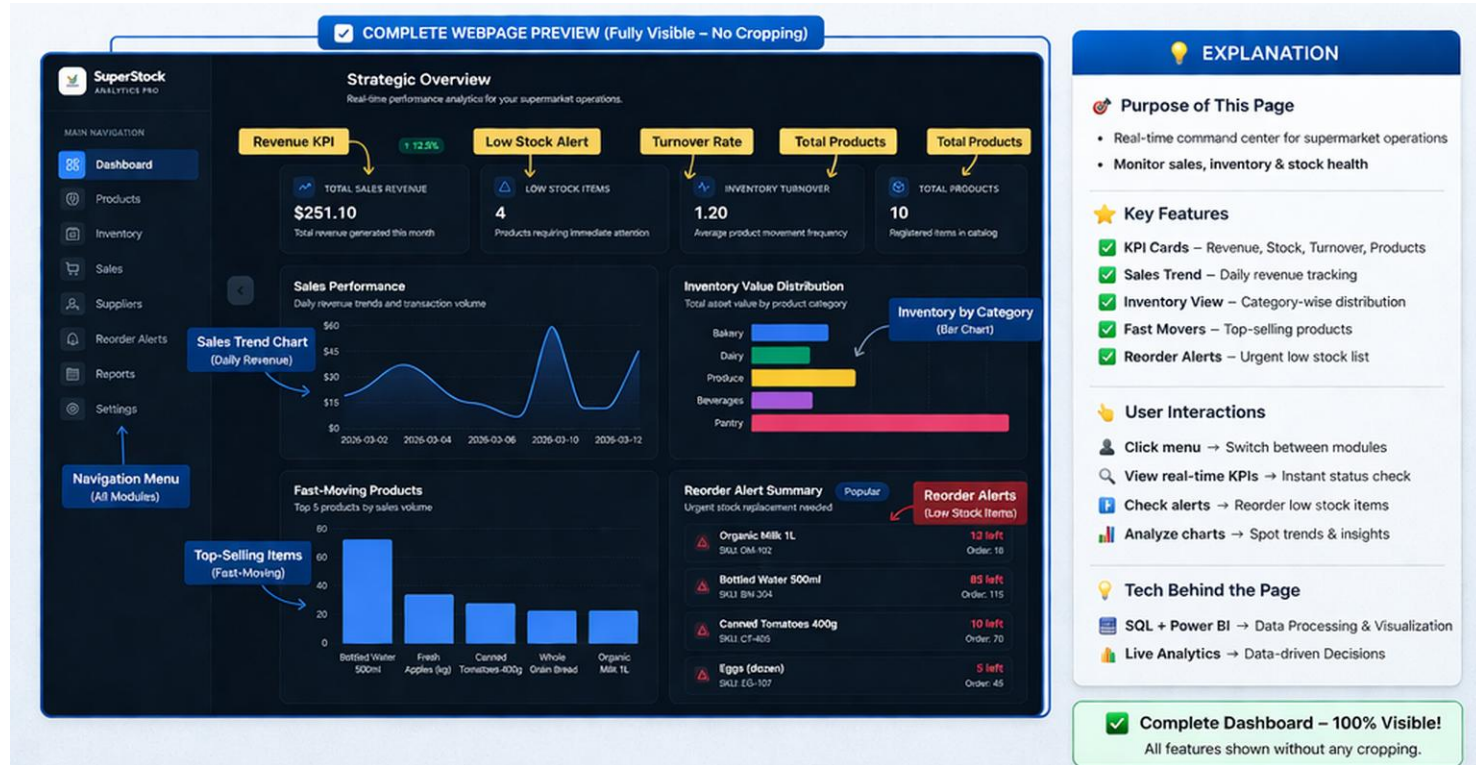
Phase 2: Power BI Modeling: * Implementing a **Star Schema** (Fact table for Sales, Dimension tables for Products/Date).

Writing **DAX Measures** for "Year-over-Year" growth and "Running Totals."

SOFTWARE ROLE



DASHBOARD (STRATEGIC OVERVIEW)



PRODUCT CATALOG

The screenshot shows the 'Product Catalog' page in 'SuperStock ANALYTICS PRO'. The interface includes a left sidebar with navigation links: Dashboard, Products, Inventory, Sales, Suppliers, Reorder Alerts, Repo, Settings, and another Settings link. The main content area displays a table of products with columns for SKU, Product Name, Category, Price, Supplier, Stock, and Actions. Annotations highlight key features: 'Search & Filter Find Products Entry' points to the search bar; '+ Add Product Button Quick Add New Stock' points to the top right button; 'Navigation Sidebar Module Navigation' points to the sidebar; 'Add Product Button Quick Add New Stock' points to a button in the table; and 'Product Inventory Table Detailed Item List' points to the table itself. The table shows 10 products, including Whole Grain Bread, Organic Milk 1L, Fresh Apples (kg), Bottled Water 500ml, and Canned Tomatoes 400g.

SuperStock ANALYTICS PRO
Manage your supermarket inventory items details.

MAIN NAVIGATION

- Dashboard
- Products
- Inventory
- Sales
- Suppliers
- Reorder Alerts
- Repo
- Settings
- Settings

Product Catalog

Search by name or SKU...

Search & Filter
Find Products Entry

+ Add Product
Quick Add New Stock

+ Add Product Button
Quick Add New Stock

SKU	Product Name	Category	Price	Supplier	Stock	Actions
W08-001	Whole Grain Bread	Bakery	\$5.00	Baker's Pride	46 SKU: 20	
O2-103	Organic Milk 1L	Dairy	\$5.20	Daily Dairy Co.	13 SKU: 15	
PA-102	Fresh Apples (kg)	Produce	\$4.00	Green Valley Produce	107 SKU: 15	
FA-202	Whole Grain Bread	Bakery	\$5.50	Baker's Pride	65 SKU: 10	
OP-003	Organic Milk 1L	Bakery	\$5.20	Daily Dairy Co.	62 SKU: 15	
CA-005	Fresh Apples (kg)	Produce	\$4.50	Green Valley Produce	65 SKU: 20	
C1-005	Bottled Water 500ml	Beverages	\$9.20	Pacific Beverages		
C2-205	Canned Tomatoes 400g	Pantry	\$8.50	Daily Dairy Co.		
W08-001	Whole Grain Bread	Bakery	\$5.20	Baker's Pride	10 SKU: 10	
O2-102	Organic Milk 1L	Bakery	\$2.20	Daily Dairy Co.	23 SKU: 10	
PA-205	Fresh Apples (kg)	Produce	\$4.00	Green Valley Produce	100 SKU: 15	
BA-001	Bottled Water 500ml	Beverages				

Showing 10 of 10 products

Navigation Sidebar
Module Navigation

Navigation Sidebar
Module Navigation

Add Product Button
Quick Add New Stock

Product Inventory Table
Detailed Item List

EXPLANATION

PURPOSE OF THIS PAGE

- Provides an organized list of products in the supermarket inventory
- Facilitates easy inventory management and stock updates

KEY FEATURES

- Product Inventory Table:** Displays SKU, product details (category, price/supplier)
- Search & Filter:** Quickly find products by name, SKU, or category
- + Add Product Button:** Easily add new inventory items
- Actions Column:** Edit or remove products with ease

USER INTERACTIONS

- Search or filter products in the inventory list
- Add new products by clicking the + Add Product: button
- Click edit/delete icons to modify or remove a product

IMPACT

Helps maintain optimal inventory levels and streamline product management.

INVENTORY MANAGEMENT

SuperStock ANALYTICS PRO

Inventory Management
Track and adjust stock level and adjustments.

MAIN NAVIGATION

- Dashboard
- Products
- Inventory
- Sales
- Suppliers
- Reorder Alerts
- Reports
- Settings
- Settings

HEALTHY STOCK
6 Items with sufficient stock

LOW STOCK
4 Requires attention soon

OUT OF STOCK
0

Stock Status Overview

Filter inventory...

Product	Current Stock	Status	Last Activity
Whole Grain Bread SKU: WG-001	45 OF 50 GOAL	Healthy	3/29/2026 07:18 PM
Organic Milk 1L SKU: OM-102	12 OF 15 GOAL	Low	3/29/2026 07:18 PM
Fresh Apples (kg) SKU: FA-103	50 OF 80 GOAL	Healthy	3/29/2026 07:18 PM
Bottled Water 500ml SKU: BW-104	85 OF 200 GOAL	Low	3/29/2026 07:18 PM
Canned Tomatoes 400g SKU: CT-105	10 OF 30 GOAL	Healthy	3/29/2026 07:18 PM
Greek Yogurt 500g SKU: GY-106	25 OF 30 GOAL	Low	3/29/2026 07:18 PM
Eggs (dozen) SKU: EG-107	5 OF 50 GOAL	Healthy	3/29/2026 07:18 PM
Pasta Spaghetti 500g SKU: PS-108	120 OF 100 GOAL	Healthy	3/29/2026 07:18 PM

Navigation Sidebar Manage Modules

Stock Status Indicates Quick Inventory Search

Sync Records
Data Synchronizes

EXPLANATION

PURPOSE OF THIS PAGE

- Monitors and controls stock levels in real-time
- Identifies critical inventory needs promptly

KEY FEATURES

- Stock Status Indicators:** Clearly shows healthy, low, and out of stock items
- Search & Filter:** Easily find specific inventory items
- Manage Stock Actions:** (+) Restock or adjust current stock levels
- Sync Records Button:** Synchronize inventory data with external systems

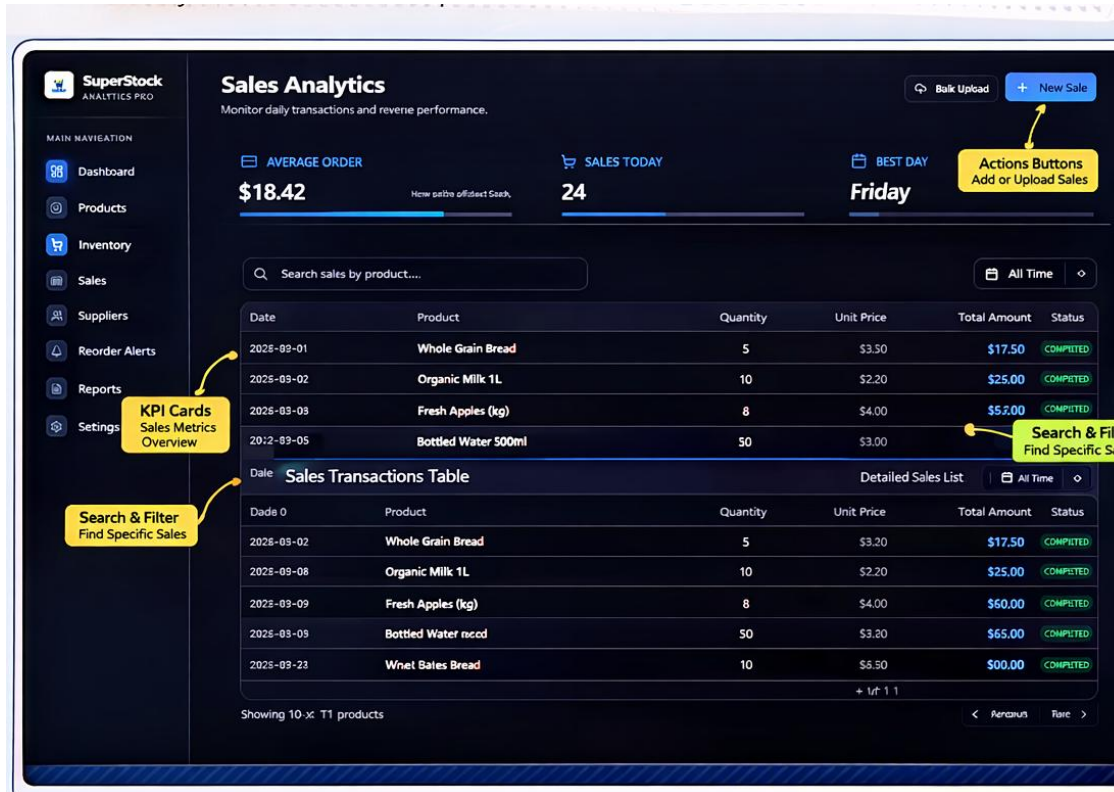
USER INTERACTIONS

- View stock status categories: Healthy, Low, Out of Stock
- Search or filter inventory for specific stock levels
- Click (+/-) to restock and adjust stock quantities

IMPACT

Helps minimize stockouts, maintain optimal stock levels, and improve order management.

SALES ANALYTICS



EXPLANATION

PURPOSE OF THIS PAGE

- Tracks daily sales transactions
- Analyzes revenue trends and performance

KEY FEATURES

- KPI Cards:** Overview of average order value, total sales, best day, orgrowth rate
- Sales Transactions Table:** Detailed list of sales showing date, product quantity, unit price and total amount.
- Search & Filter:** Find sales by product name
- Actions Buttons:** Bulk upload sales or add new orders easily.

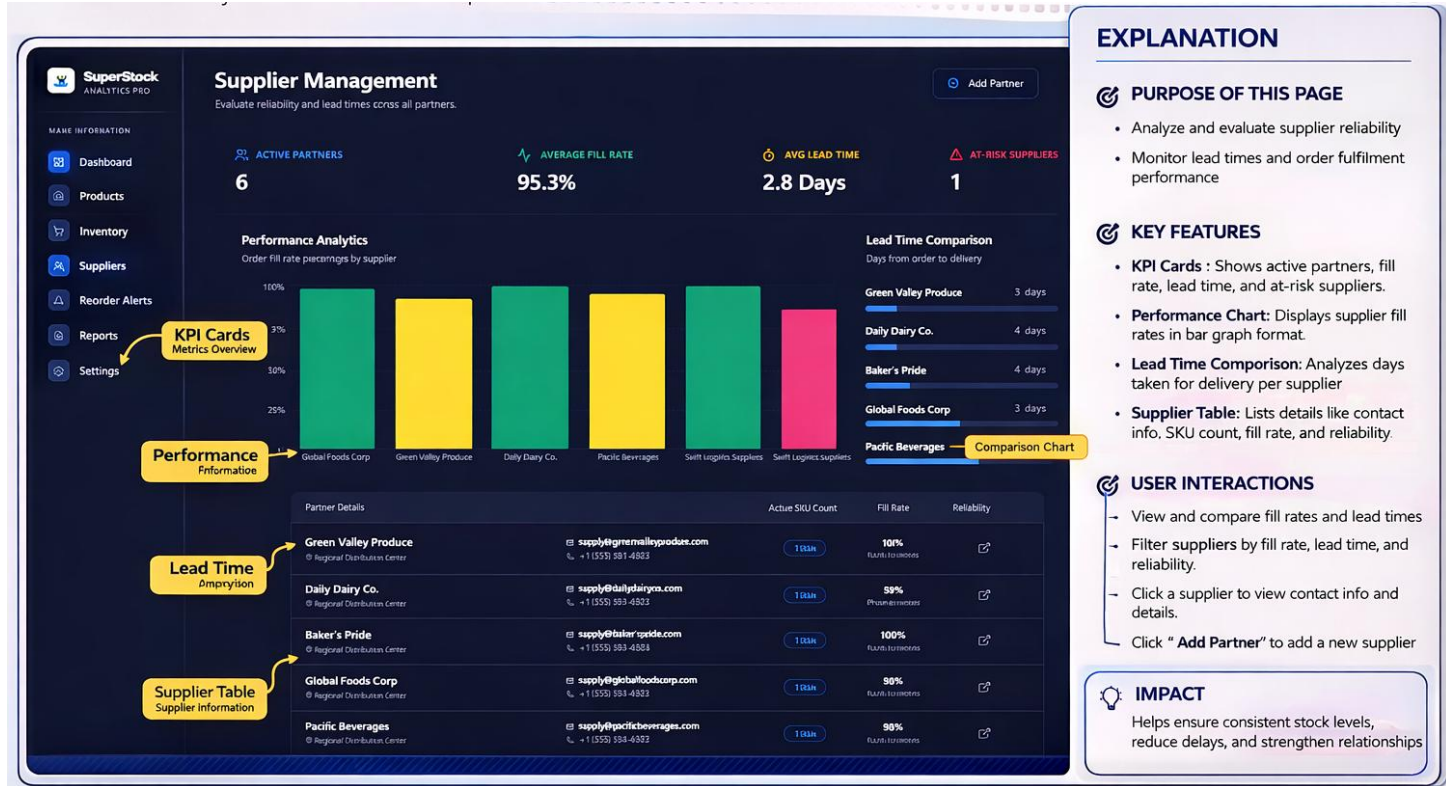
USER INTERACTIONS

- View daily sales metrics in the KPI cards
- Search for specific product sales
- Filter by date range to analyze specific periods
- Click 'New Sale' to manually input orders
- Use 'Bulk Upload' to import sales data quickly

IMPACT

Improves sales tracking - helps identify high performing products and boosts revenue growth.

SUPPLIER MANAGEMENT



REORDER MANAGEMENT

SuperStock ANALYTICS PRO

Reorder Management

Automated inventory replenishment alerts and tracking.

4 PENDING ACTIONS

Needs Order (4) **In Transit** (0) **History** (0)

Organic Milk 1L (SKU: OM-382) Critical

STOCK	SUGGESTED
12	+18

Supplier: Daily Dairy Co.
Lead Time: 1 Days

Generate Order

No active orders
Track incoming shipments here.

Order History
Completed reorders will appear here.

Needs Order (Low Stock Alerts) **Generate Order**

Pending Actions (4) **Summary count**

Organic Milk 1L (SKU: OM-382)

STOCK	SUGGESTED	STOCK ITEMS
12	+18	+115

Supplier: Daily Dairy Co.
Lead Time: 1 Days

Generate Order **Summary Count** **In Transit** **Order History**

Canned Tomatoes 400g (SKU: CT-485) Critical

STOCK	SUGGESTED
40	+70

EXPLANATION

PURPOSE OF THIS PAGE

- Automates inventory reordering
- Tracks reorder process in one place.

KEY FEATURES

- Needs Order:** Shows items needing immediate reordering.
- Pending Actions:** Displays number of pending reorder tasks.
- In Transit:** Tracks incoming shipments
- History:** Records completed reorder transactions.
- Generate Order:** Quickly create new purchase orders.

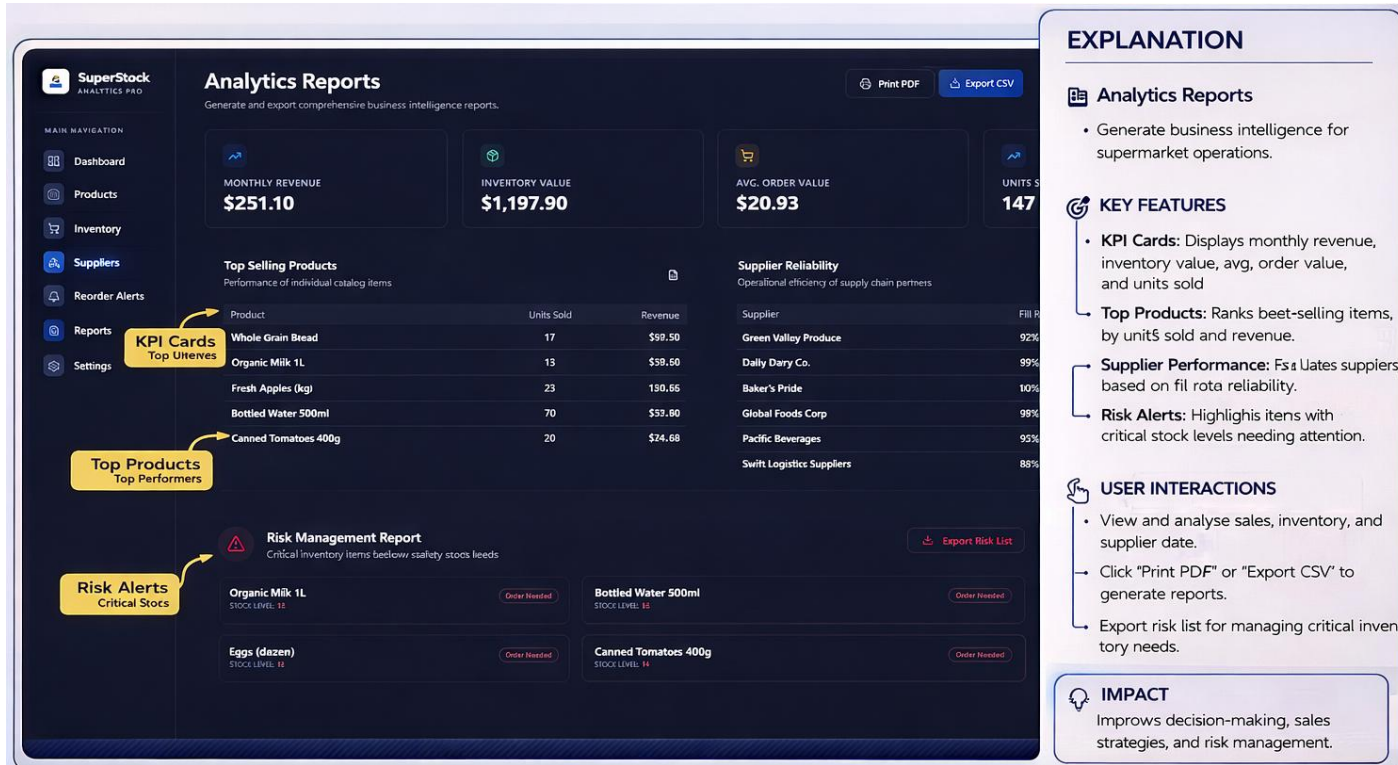
USER INTERACTIONS

- Review low stock alerts for items needing restocking.
- Click 'Generate Order' to quickly create a reorder.
- Track shipments and view reorder history in separate sections.

IMPACT

Ensures timely restocking, prevents stockouts, maintains inventory efficiency.

REPORTS & ANALYTICS



SYSTEM SETTINGS

System Settings
Configure your analytics system rules or preferences.

Application
Modify the visual and interactive behavior of the dashboard.

- General
- Notifications
- Security
- Data Management

Inventory Rules
Define Inventory Alerts

Inventory Calculation Rules
Define how the system identifies stock risks and suggests reorders.

Default Reorder Level
The baseline stock count that triggers an alert.
20

Automatic Reorder Suggestions
Automatically calculate quantities based on lead times. ☒

Low Stock Email Alerts
Notify the warehouse manager via email on critical stock. ☒

Danger Zone
Reset All Data

Reset Application Data
Restore all catalog, sales, and inventory or initial mock data. [Reset Demo](#)

SuperStock Strategic Analytics v1.0.4
Developed by Mlinda. This is a demonstration environment with no persistent data.
● SYSTEM HEALTHY ● API CONNECTED ● 2025 READY

EXPLANATION

PURPOSE OF THIS PAGE

- Customize the analytics system
- Adjust alert levels and preferences.

KEY FEATURES

- System Configuration:** Control navigation, notifications, and data.
- Inventory Rules:** Define reorder and safety stock thresholds.
- Danger Zone:** Reset all application data.
- Version Info:** Displays current software version.

USER INTERACTIONS

- Navigate and modify system settings.
- Adjust inventory reorder rules.
- Toggle email alerts for low stock.
- Reset all data as irreversible action.

IMPACT

Tailors analytics platform to specific operational needs.

RESULT ANALYSIS

- **Inventory Turnover Ratio (ITR):** How many times a supermarket sells and replaces its stock in a period.
- **Stock-to-Sales Ratio:** Comparing average inventory on hand to net sales.
- **Gross Margin Return on Investment (GMROI):** Proving which items are actually making the most profit per dollar spent on stock.
- **Out-of-Stock (OOS) Rate:** Tracking the frequency of empty shelves.

CONCLUSION

The Supermarket Analytics project successfully proves that combining **SQL's data processing power** with **Power BI's visualization capabilities** creates a high-value business tool. This system allows managers to move away from "guessing" and toward "data-driven" ordering. It ensures the right product is on the right shelf at the right time, maximizing both profit and customer loyalty.

REFERENCES

S.No	Source Title	Author / Publisher	Category
01	<i>SQL for Data Analytics</i>	Upom Malik (Packt Publishing)	Technical Book 
02	<i>The Definitive Guide to DAX</i>	Alberto Ferrari & Marco Russo	Reference Guide 
03	"Inventory Management in Large Scale Retail"	IEEE Xplore Journal	Academic Paper 
04	Power BI Learning Path for Data Analysts	Microsoft Documentation	Official Technical Docs 
05	Storytelling with Data	Cole Nussbaumer Knaflic	Visualization Theory 

THANK YOU